

Wilson Costa

Lisbon, Portugal

Portfolio: wilsoncosta-portfolio.vercel.app | Github: github.com/WilsonRCosta | LinkedIn: linkedin.com/in/wilson-costa-9a6b96159
wilson.ruben97@gmail.com | (+351) 932 318 736

Professional Experience

BMW - Critical Techworks - Devoteam

February 2023 – November 2025

Backend Engineer and Scrum Master

- Served as **Backend Engineer and Scrum Master** on a large-scale distributed platform built with **Quarkus microservices**, each owning its own **PostgreSQL schema**, deployed on **Kubernetes**, with CI/CD via **GitHub Actions** and infrastructure provisioning using **Terraform & Terragrunt (Azure)**.
- Designed and developed a **big-data backend system** to generate and manage vehicle variants for powertrain data, scaling to **millions of variant-feature value assignments** across services.
- Led a **Kafka-based data migration**, creating a dedicated microservice to process thousands of ~2MB JSON events, transforming slow batch workflows into an **asynchronous near-real-time pipeline** using partitioning and parallelization.
- Optimized **monolith import pipelines**, improving Hibernate queries and in-memory graph traversal, reducing processing time from **~20 minutes to ~15 seconds**.
- Implemented **scheduled processing jobs and complex business logic** based on logical expressions, enabling users to dynamically select subsets of variants without generating all possible combinations.
- Designed and optimized **PostgreSQL schemas with advanced table partitioning**, query rewrites, and indexing, reducing query latency to **~500ms** on **10M+ row partitioned tables**.
- Built and optimized **feature-value assignment engines and user-driven workviews**, ensuring scalability and responsive UX through efficient data-access patterns and default filtering.
- Improved **code quality, maintainability, and delivery workflows**, using **SonarQube**, contributing to architectural decisions, collaborating with cross-team stakeholders, and supporting CI/CD and production operations.

BMW - Critical Techworks - Devoteam

October 2021 - February 2023

Full Stack Developer

- Delivered fullstack features for a modular widget-based architecture with a dashboard platform with **real-time data** and third-party integrations.
- Built Quarkus and NestJS APIs, exposing data via GraphQL, CosmosDB (cloud NoSQL) and PostgreSQL for data persistence.
- Managed Kubernetes deployments via Helmfile, Jenkins and ArgoCD pipelines for automated builds and releases.
- **Built Widget Ecosystem Development:** Built modular, reusable widgets (e.g. sales timelines, world clock, car descriptions, visualization of car errors, and Microsoft Teams integration) supported by APIs, improving dashboard extensibility.
- **Integration of a Content Management System:** Implemented a news side panel integrated with Strapi as the CMS, enabling designated users to create and manage news articles, which are dynamically displayed in the side panel.
- **Designed High-Performance APIs:** The backend microservices achieved **<1 second response times**, enabling real-time widget updates.
- **Database Engineering:** Developed and optimized queries in **CosmosDB and PostgreSQL**, improving data reliability and supporting high-traffic dashboard usage for **1000+** users.
- **Frontend Migration Update:** Applied an Angular migration from **version 12 to 14** and a BMW internal library using the latest components and styles.

- Worked on modernizing legacy **Oracle Forms applications** into web-based systems, improving maintainability, performance, and frontend-backend integration.
- Used AngularJS for a **legacy modernization** of user experience and an interactive web UI. Groovy on Grails for server-side logic. Oracle SQL for data persistence. GitLab and Jenkins for source control and automated deployments.
- **Data Modernization & Integration:** Recreated the **data creation workflow using JSON**, enabling seamless communication with AngularJS front-end.
- **Business Logic Optimization:** Refactored legacy **JavaScript tax functions** to be simpler, more efficient, and reliable.
- **Performance Improvements:** Reviewed auto-transformed APIs and **optimized database access**, significantly reducing query times and improving system responsiveness.

Education

Instituto Superior de Engenharia, Lisbon (ISEL)

Master's Degree in Informatics and Computer Engineering (only 1st year)

September 2021

Focus: Advanced Software Engineering, Distributed Systems, Cybersecurity

Main Projects:

- *Travel Planner App* + *API* (Software Engineering): Angular, Spring Boot - Java, PostgreSQL, Kubernetes Deployments, Google Authentication, Trip Advisor API
- *CES Vector* (Distributed Systems): Transaction management system (Quarkus, Kubernetes, Docker)

Bachelor's Degree in Informatics and Computer Engineering

October 2020

Focus: Programming, Web Development, Information Systems, Cloud Computing, Security, Networking

Main Projects:

- *Invoice Manager* (BSc Final Project): Laravel API + React frontend with OCR integration using Tesseract
- *Text System* (Cloud Computing): Java-based system on GCP using Vision & Translation APIs with dynamic VM allocation
- *Project Manager* (Web Development): React + Spring Boot - Kotlin + PostgreSQL
- *Ciborg Games* (Internet Programming): Javascript + ExpressJS (Node with Auth)

Skills

Programming Languages: Java, TypeScript, JavaScript, SQL, Kotlin, NodeJS

Frameworks & Tools: Quarkus, Spring Boot, Angular, React, Kafka, Helmfile, Terraform, Terragrunt, Kubernetes, Jenkins, GitHub Actions

Databases: PostgreSQL, MySQL, MongoDB, Elasticsearch, CosmosDB, Oracle SQL

Spoken Languages: **Portuguese** (native), **English** (fluent), Spanish (basic)

Certifications:

- Professional Scrum Master – Scrum.org | 2024
- AWS Certified Cloud Practitioner – Amazon Web Services | 2022
- Associate Cloud Engineer – Google Cloud | 2021

Personal Projects:

- *Shining Code* (Live Demo): Fashion E-Commerce Web App.